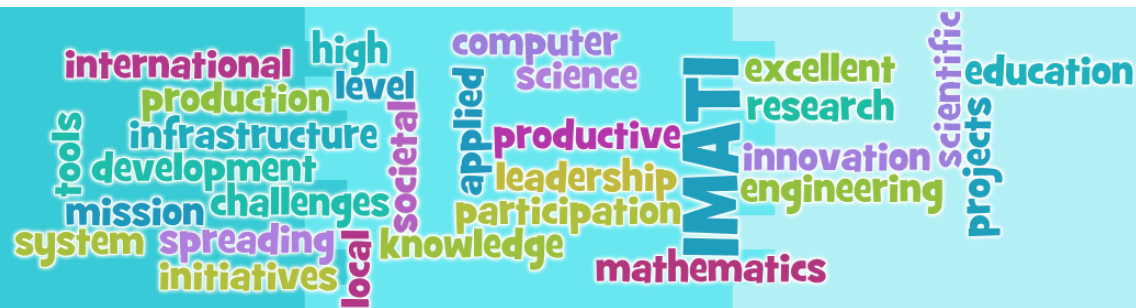


Extracting Geometrical Features From Data

Topological Data Analysis

Ulderico Fugacci

CNR - IMATI



Topological Data Analysis

Outline:

The Notion of Shape

Simplicial Complexes

Simplicial Homology

From Data to Complexes

Persistent Homology

Visualizing Persistence

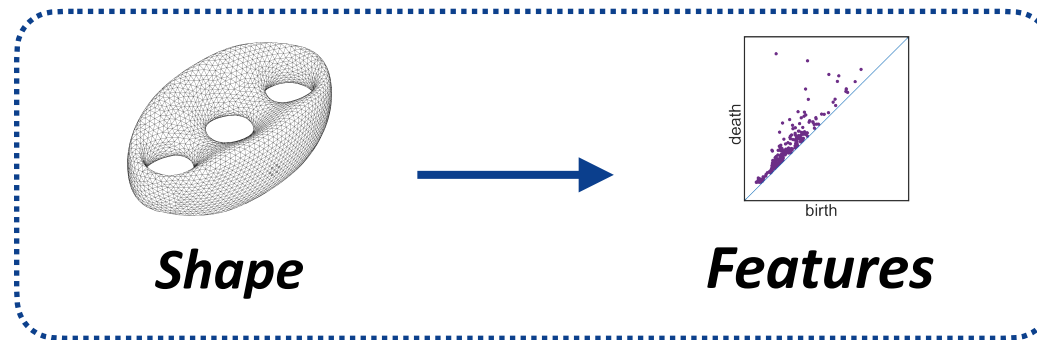
Persistence & Stability

Computing Persistence

Visualizing Persistence

Persistent Homology

(Persistent) Homology allows for assigning to any (filtered) simplicial complex *topological information expressed in terms of algebraic structures*



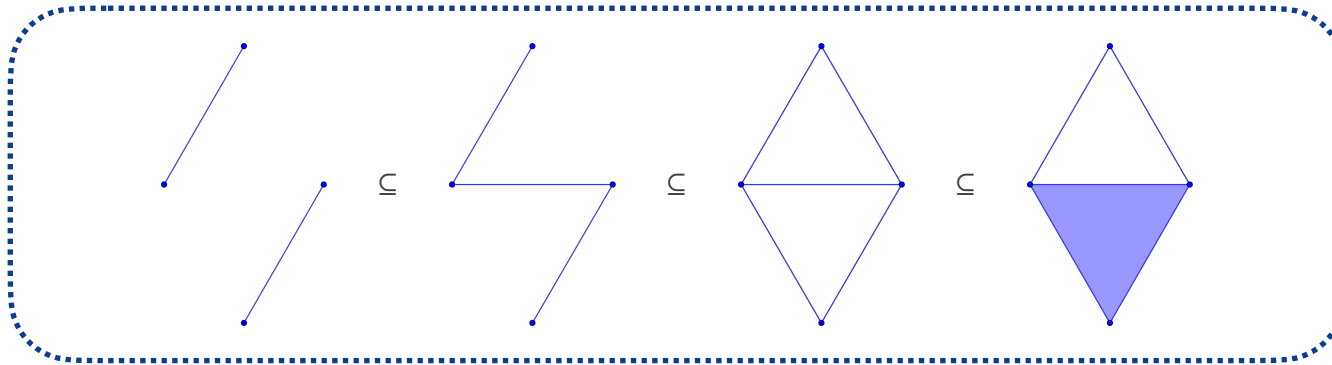
Goal:

We address two main questions:

- ♦ Can this topological information be *characterized in a simpler and “more visualizable” way?*
- ♦ Is this information *stable under small perturbations of the input data?*

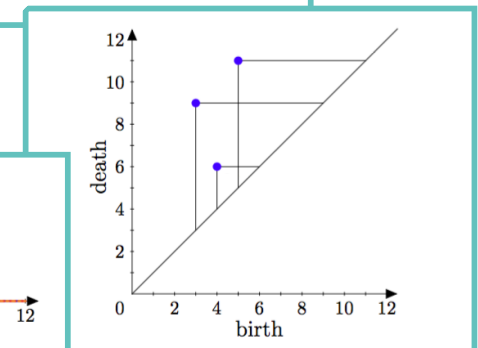
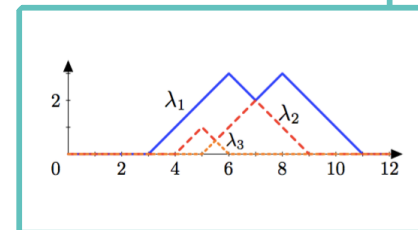
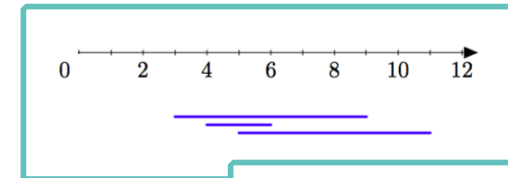
Visualizing Persistence

Given a filtration $\mathcal{F}$,



Persistent pairs of $\mathcal{F}$ can be visualized through:

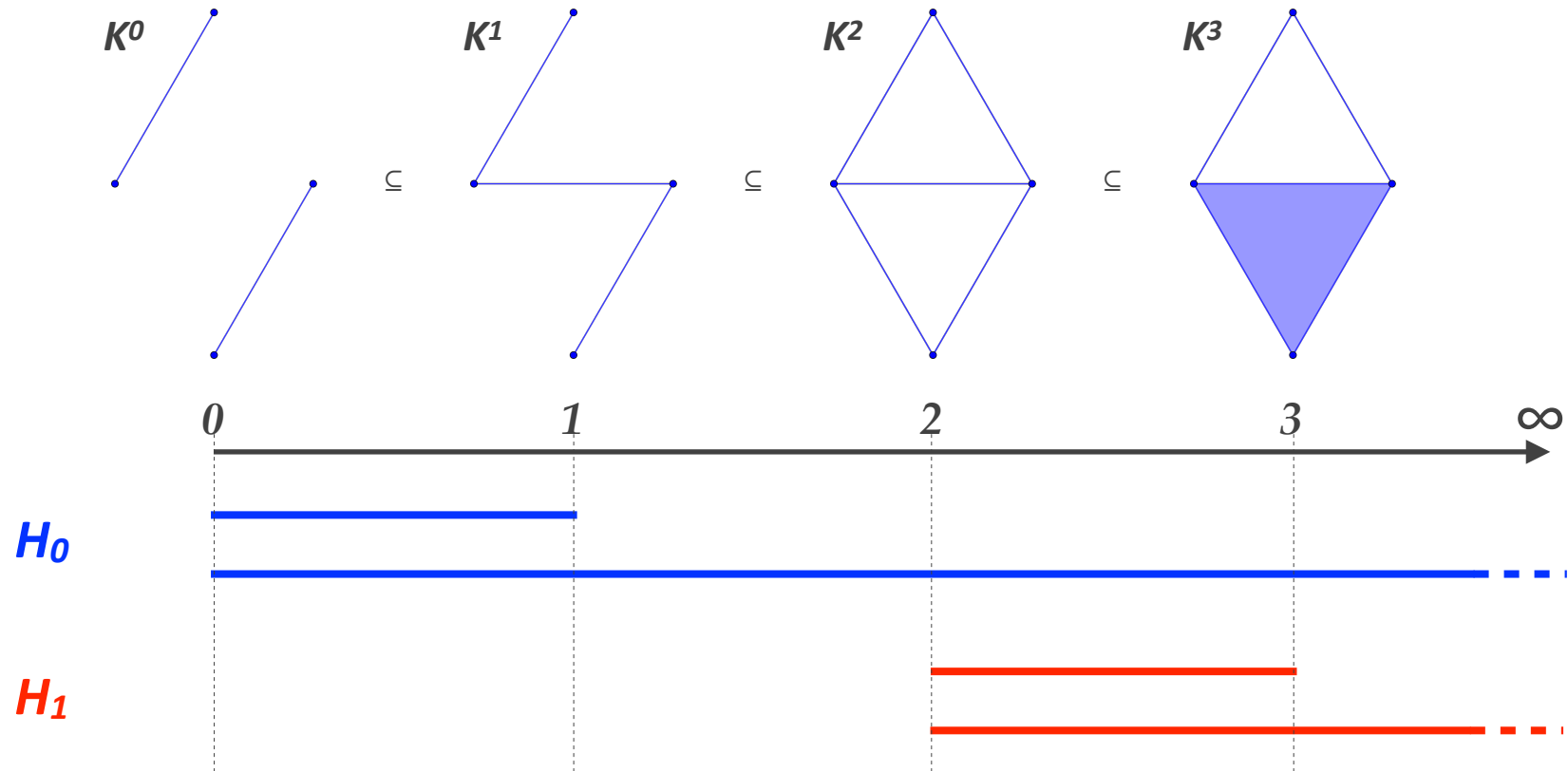
- ◆ **Barcodes** [Carlsson et al. 2005; Ghrist 2008]
- ◆ **Persistence diagrams** [Edelsbrunner, Harer 2008]
- ◆ **Persistence landscapes** [Bubenik 2015]
- ◆ **Corner points and lines** [Frosini, Landi 2001]
- ◆ **Half-open intervals** [Edelsbrunner et al. 2002]
- ◆ **k -triangles** [Edelsbrunner et al. 2002]



Visualizing Persistence

Barcodes:

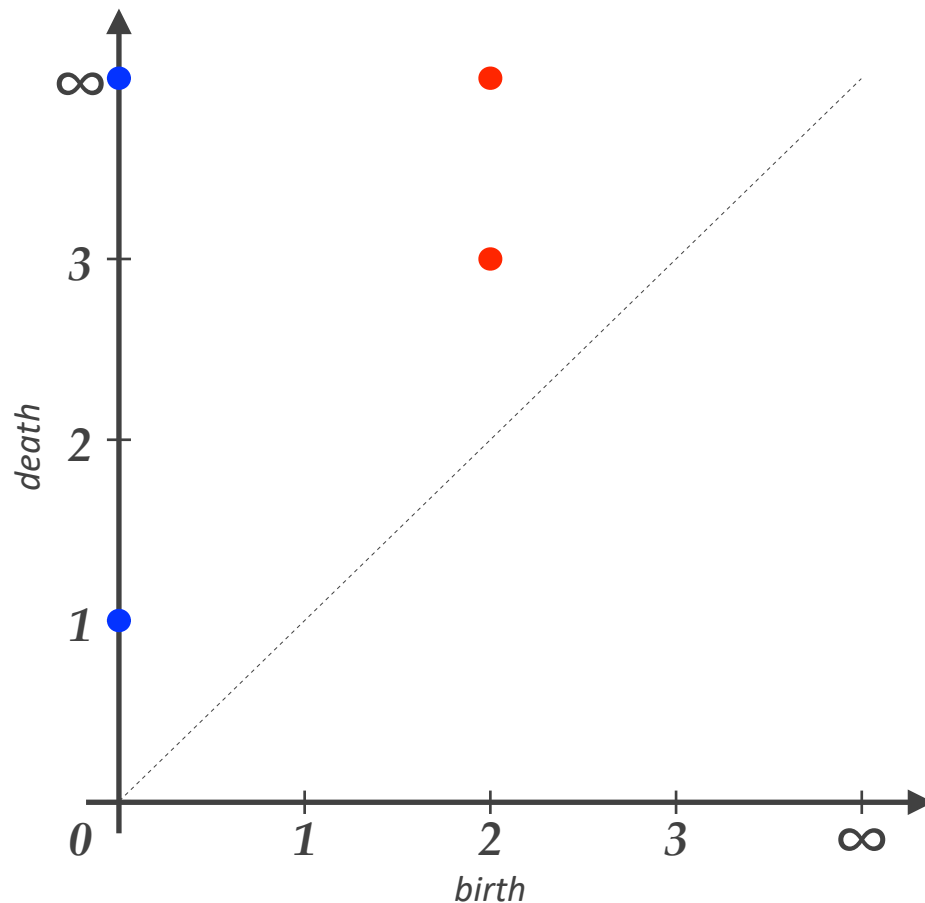
Persistence pairs are represented as *intervals* in $\mathbb{R}$



Visualizing Persistence

Persistence Diagrams:

Persistence pairs are represented as *points in $\mathbb{R}^2$*



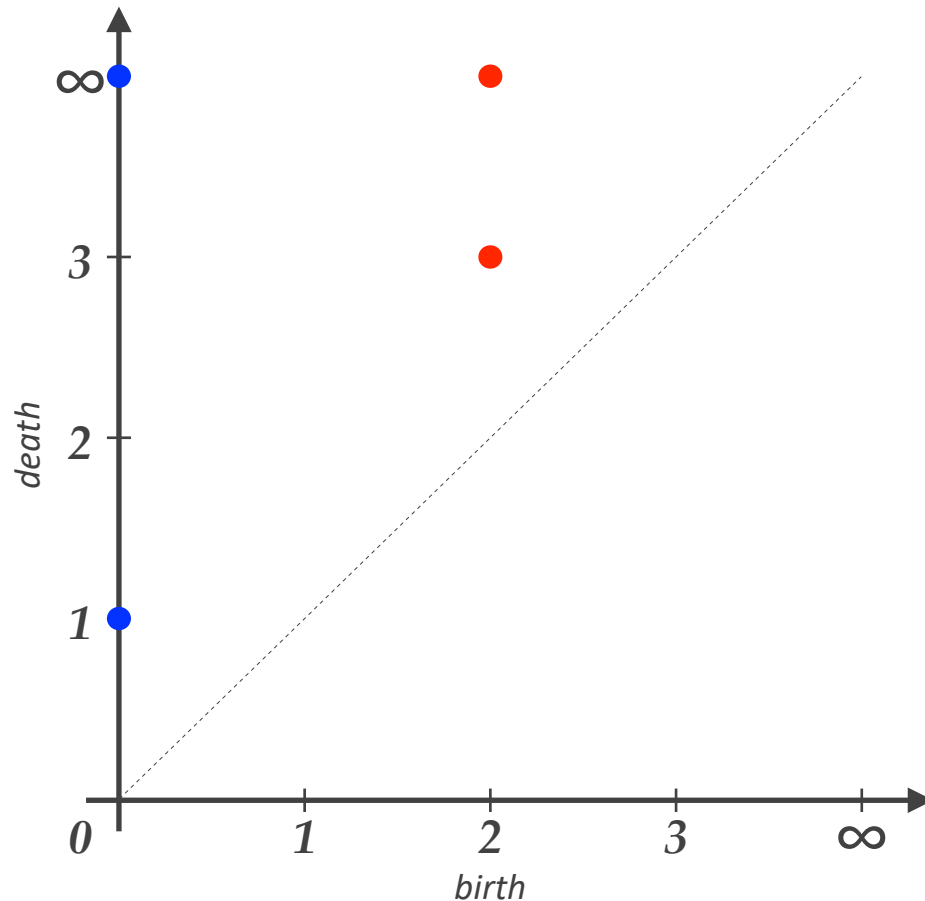
H_0 $(0, 1)$
 $(0, \infty)$

H_1 $(2, 3)$
 $(2, \infty)$

Visualizing Persistence

Persistence Diagrams:

Persistence pairs are represented as **points in $\mathbb{R} \times (\mathbb{R} \cup \{\infty\})$**



H_0 $(0, 1)$
 $(0, \infty)$

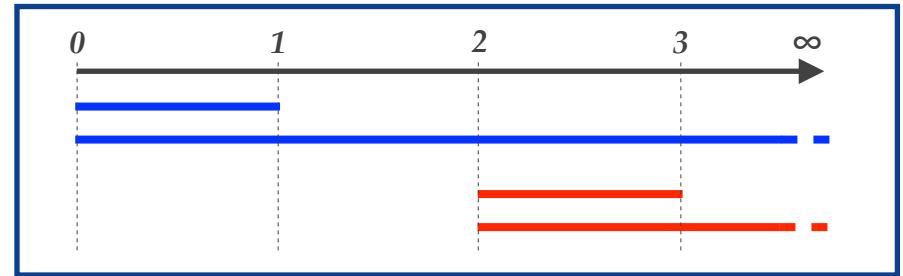
H_1 $(2, 3)$
 $(2, \infty)$

Formally, a persistence diagram is a **multiset**

- ♦ Points are endowed with **multiplicity**

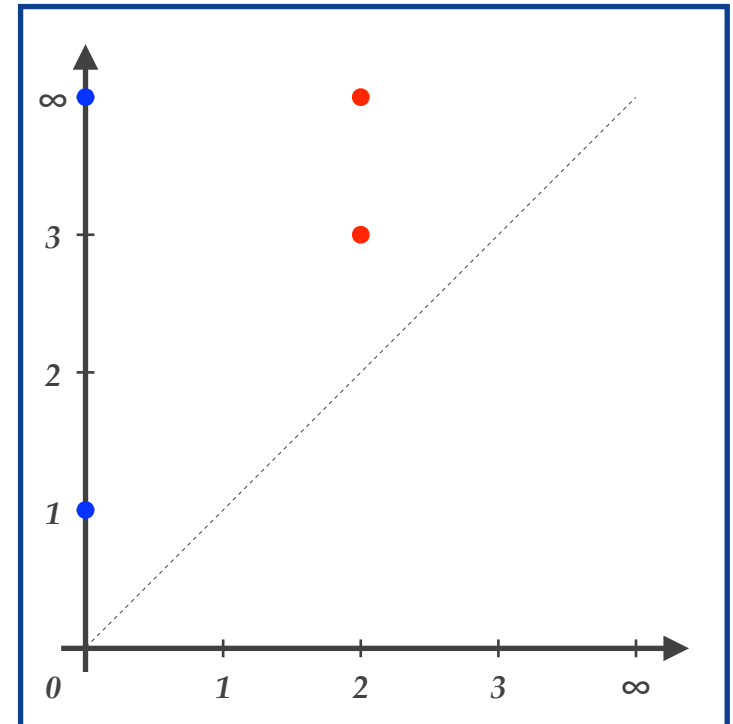
Visualizing Persistence

Both tools **visually represent** the **lifespan** of the homology classes:



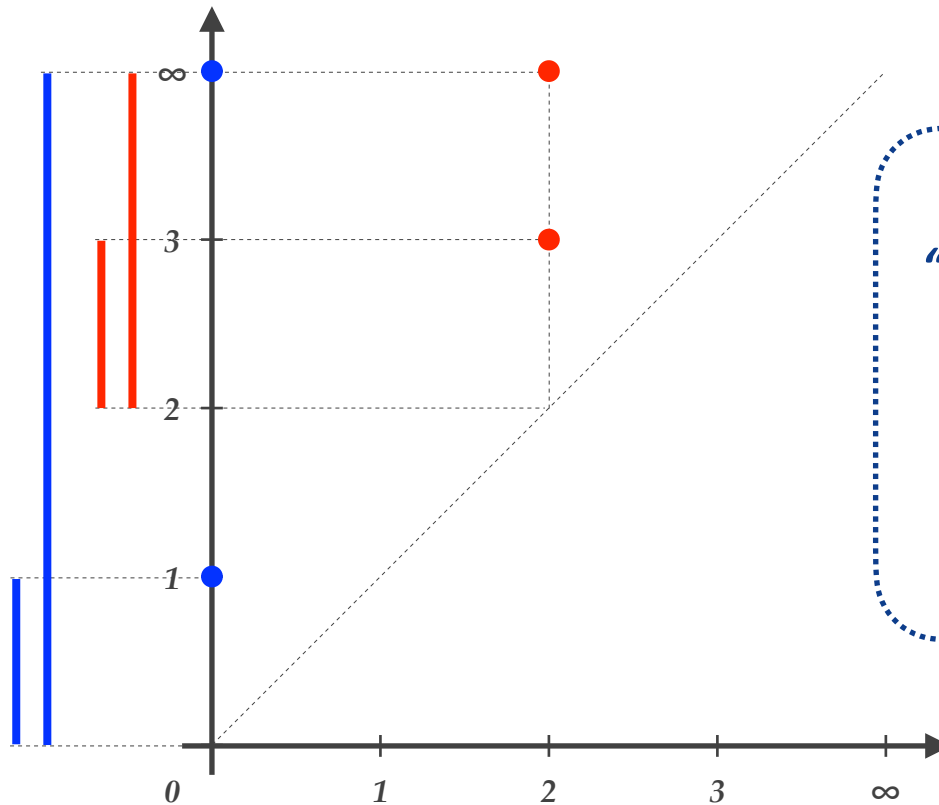
- ◆ Barcode: **length of the intervals**
- ◆ Persistence Diagram: **distance from the diagonal**

Barcodes and Persistence Diagrams
encode equivalent information



Visualizing Persistence

Barcodes and Persistence Diagrams *encode equivalent information*



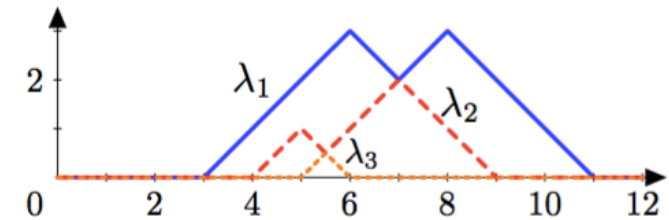
A visualization can be easily
“translated” into the other one:

$$\begin{array}{ccc}
 [p, q] & & (p, q) \\
 [p, \infty) & \longleftrightarrow & (p, \infty)
 \end{array}$$

Visualizing Persistence

Persistence Landscapes:

Persistence landscapes are statistics-friendly representations of persistence pairs



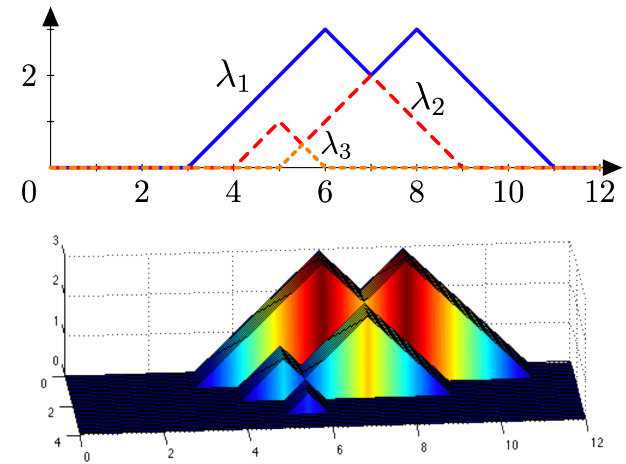
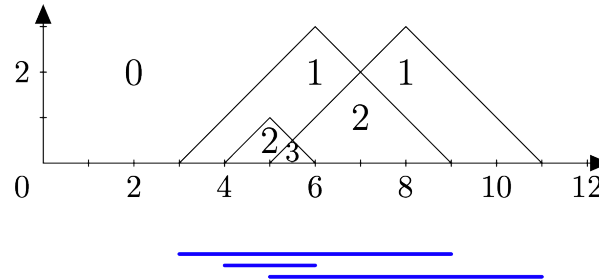
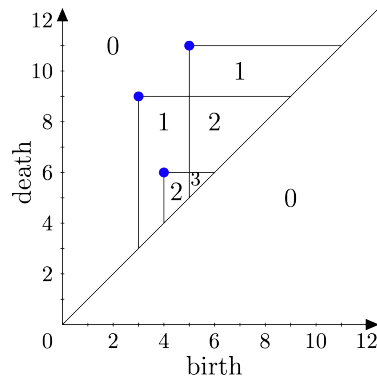
Given a persistence module M , persistence landscapes

- ♦ Consist of a collection of **1-Lipschitz functions**
- ♦ Lie in a **vector space**
- ♦ Are **stable** (under small perturbations of the input filtration)

Visualizing Persistence

Persistence Landscapes:

Given a persistence module M ,



Formally,

$$\lambda_i(x) := \sup\{m \geq 0 \mid \beta^{x-m, x+m} \geq i\}$$

$$\text{where } \beta^{p,q} := \dim(\text{Im}(\mu^{p,q} : M_p \rightarrow M_q))$$

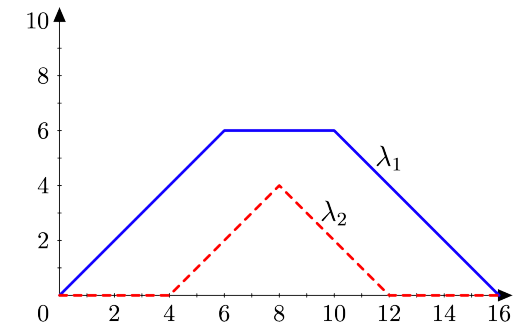
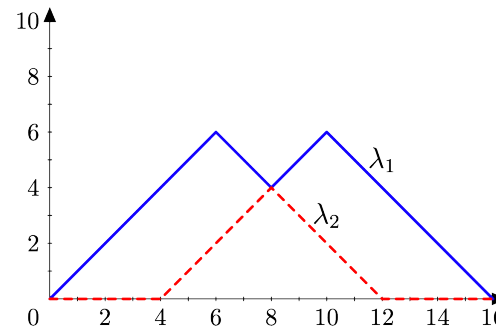
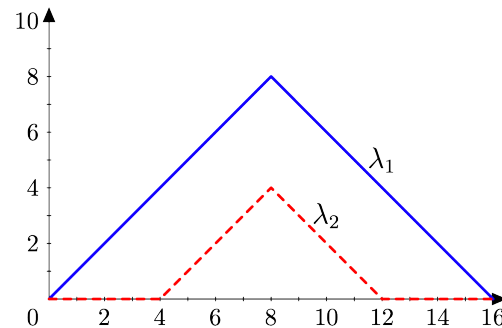
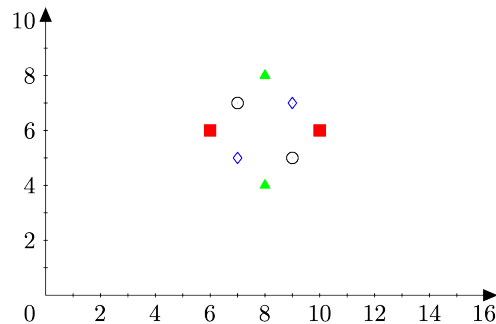
Images from [Bubenik 2015]

Visualizing Persistence

Persistence Landscapes:

*Mean of persistence diagrams is **not unique**, but ...*

*Mean of persistence landscapes is **well-defined***



Images from [Bubenik 2015]

Bibliography

Some References:

- ♦ **Persistent Homology:**
 - ✦ U. Fugacci, S. Scaramuccia, F. Iuricich, L. De Floriani. ***Persistent homology: a step-by-step introduction for newcomers***. Eurographics Italian Chapter Conference, pages 1-10, 2016.

Persistence & Stability

Stability of Persistence

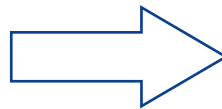
In order to be adopted in real applicative domains, it is crucial that

persistent homology is not affected by noisy data and small perturbations

Stability Result:

By defining **distances*** for both domains,

Similar Data



**Similar
Persistent Homology**

**The term “distance” is intended in a broad sense, including pseudo-metrics and dissimilarity measures*

Stability of Persistence

Distances:

♦ *For the Data in Input:*

- ❖ *Natural pseudo-distance* of shapes
- ❖ *L_∞ -distance* of filtering functions
- ❖ *Gromov-Hausdorff distance* of metric spaces/point clouds

♦ *For the Retrieved Persistent Homology Information:*

- ❖ *Interleaving distance* of persistence modules
- ❖ *Bottleneck (a.k.a. Matching) distance* of persistence diagrams
- ❖ *Hausdorff distance* of persistence diagrams
- ❖ *Wasserstein distances* of persistence diagrams

Stability of Persistence

Distances for Input Data:

Let (X, f) be a **pair** such that:

- ♦ X is a **(triangulable) topological space**
- ♦ $f: X \rightarrow \mathbb{R}$ is a **continuous function**

A pair (X, f) induces a **filtration**:

- ♦ $X^t := f^{-1}((-\infty, t])$

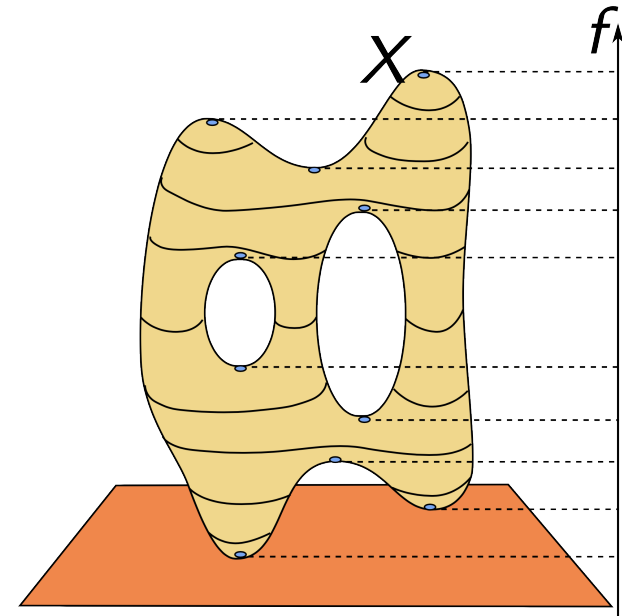


Image from [Ferri et al. 2015]

Definition:

The function f is called **tame** if:

- ♦ f has a **finite number of homological critical values** (i.e. the “time” steps in which homology changes)
- ♦ For any $k \in \mathbb{N}$ and $t \in \mathbb{R}$, the **homology group $H_k(X^t, \mathbb{F})$ has finite dimension**

Stability of Persistence

Distances for Input Data:

Definition:

Given two pairs (X, f) and (Y, g) , their *natural pseudo-distance* d_N is defined as:

$$d_N\left((X, f), (Y, g)\right) := \begin{cases} \inf_{h \in H(X, Y)} \{\max_{x \in X} \{|f(x) - g \circ h(x)|\}\} \\ +\infty & \text{if } H(X, Y) = \emptyset \end{cases}$$

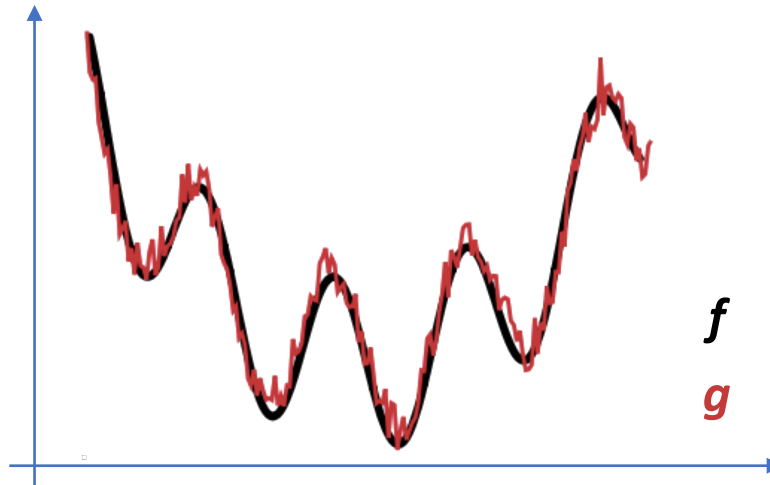
where $H(X, Y)$ is the set of all the *homeomorphisms between X and Y*

Stability of Persistence

Distances for Input Data:

Working with two functions $f, g: X \rightarrow \mathbb{R}$ defined on the same topological space X , one can simply consider the L_∞ -distance between f and g

$$\|f - g\|_\infty := \sup_{x \in X} \{|f(x) - g(x)|\}$$



Stability of Persistence

Distances for Input Data:

Given two **finite metric spaces** (X, d_X) , (Y, d_Y) (e.g. two finite point clouds in $\mathbb{R}^n$),

Definitions:

A **correspondence** $C: X \rightrightarrows Y$ from X to Y is a subset of $X \times Y$ such that the **canonical projections** $\pi_X: C \rightarrow X$ and $\pi_Y: C \rightarrow Y$ are both **surjective**

The **distortion** $\text{dis}(C)$ of a correspondence $C: X \rightrightarrows Y$ is defined as:

$$\text{dis}(C) := \sup \{ |d_X(x, x') - d_Y(y, y')| : (x, y), (x', y') \in C \}$$

The **Gromov-Hausdorff distance** d_{GH} between (X, d_X) and (Y, d_Y) is defined as:

$$d_{GH}(X, Y) := \frac{1}{2} \inf \{ \text{dis}(C) \mid C: X \rightrightarrows Y \text{ is a correspondence} \}$$

Stability of Persistence

Distances for Persistent Homology Information:

Two persistence modules M and N are called **ε -interleaved** with $\varepsilon \geq 0$ if there exist f and g such that, for any $p, q \in \mathbb{R}$ with $p \leq q$, the following **diagrams commute**

$$\begin{array}{ccc}
 & M_p & \\
 g_{p-\varepsilon} \nearrow & & \searrow f_p \\
 N_{p-\varepsilon} & \longrightarrow & N_{p+\varepsilon}
 \end{array}
 \qquad
 \begin{array}{ccccc}
 M_p & \longrightarrow & M_q & & \\
 & \searrow f_p & & \searrow f_q & \\
 & N_{p+\varepsilon} & \longrightarrow & N_{q+\varepsilon} &
 \end{array}$$

$$\begin{array}{ccc}
 M_{p-\varepsilon} & \longrightarrow & M_{p+\varepsilon} \\
 & \searrow f_{p-\varepsilon} & \nearrow g_p \\
 & N_p &
 \end{array}
 \qquad
 \begin{array}{ccccc}
 & M_{p+\varepsilon} & \longrightarrow & M_{q+\varepsilon} & \\
 g_p \nearrow & & & \nearrow g_q & \\
 N_p & \longrightarrow & N_q & &
 \end{array}$$

Definition:

Given two persistence modules M and N , their **interleaving distance** d_I is defined as:

$$d_I(M, N) := \inf\{\varepsilon \geq 0 \mid M \text{ and } N \text{ are } \varepsilon\text{-interleaved}\}$$

Stability of Persistence

Distances for Persistent Homology Information:

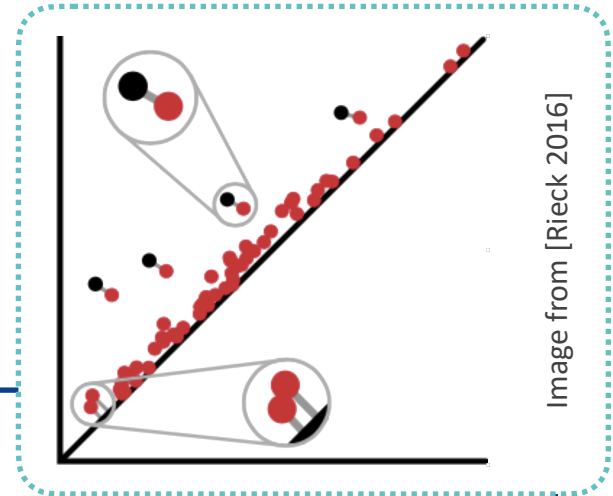
Definitions:

Given two persistence diagrams D_1 and D_2 , their **bottleneck distance** d_B and **Hausdorff distance** d_H are defined as:

$$d_B(D_1, D_2) := \inf_{\gamma} \left\{ \sup_{x \in D_1} \{ \|x - \gamma(x)\|_{\infty} \} \right\}$$

$$d_H(D_1, D_2) := \max \left\{ \sup_{x \in D_1} \left\{ \inf_{y \in D_2} \{ \|x - y\|_{\infty} \} \right\}, \sup_{y \in D_2} \left\{ \inf_{x \in D_1} \{ \|y - x\|_{\infty} \} \right\} \right\}$$

where γ ranges over all bijections from D_1 to D_2



Stability of Persistence

Distances for Persistent Homology Information:

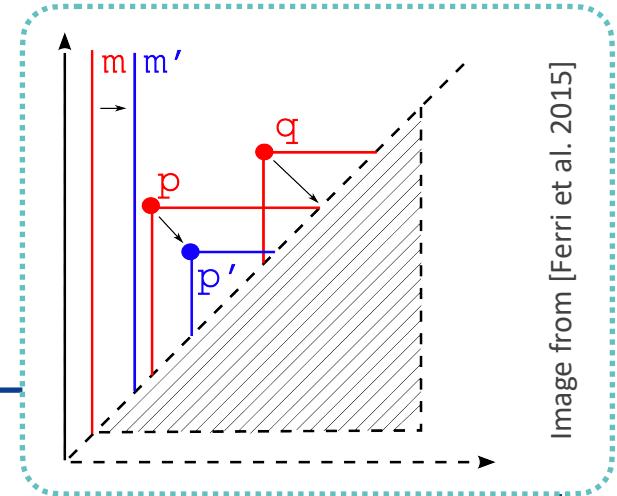
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where γ ranges over all bijections from D_1 to D_2



Stability of Persistence

Stability Results:

Given two pairs (X, f) , (Y, g) of topological spaces and **tame** functions and $k \in \mathbb{N}$, let M , N be the induced k^{th} persistence modules and let D_1 , D_2 be the corresponding persistence diagrams

- ♦ $d_H(D_1, D_2) \leq d_B(D_1, D_2)$
- ♦ $d_I(M, N) = d_B(D_1, D_2)$

Theorem:

Under the above hypothesis, the following **optimal lower bound** holds

$$d_I(M, N) \leq d_N \left((X, f), (Y, g) \right)$$

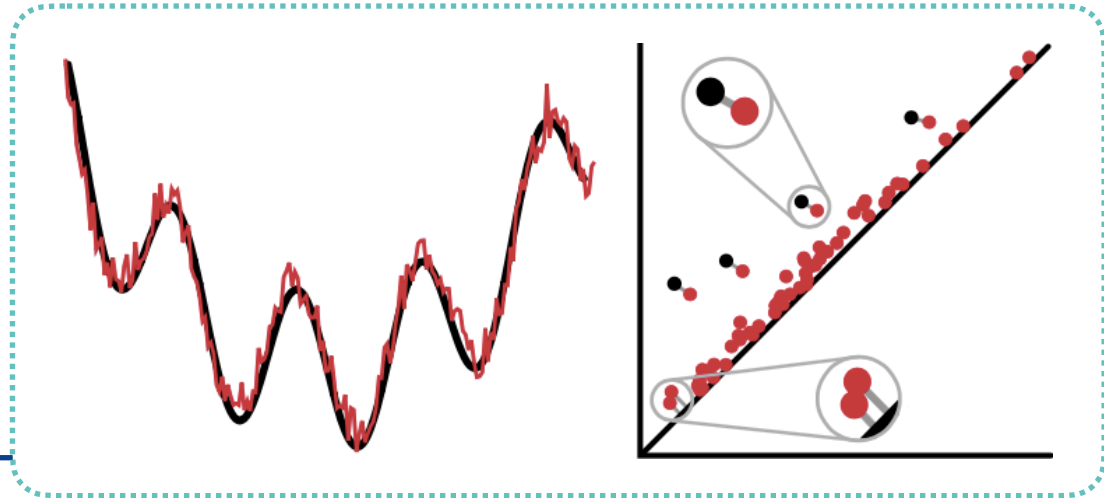
Stability of Persistence

Stability Results:

Theorem:

Given two **tame** continuous functions $f, g: X \rightarrow \mathbb{R}$
on a topological space X , $k \in \mathbb{N}$, and D_f, D_g the induced k^{th} persistence diagrams,

$$d_B(D_f, D_g) \leq \|f - g\|_\infty$$



Stability of Persistence

Stability Results:

Theorem:

Given two finite metric spaces (X, d_X) , (Y, d_Y) , $k \in \mathbb{N}$, and D_X, D_Y the k^{th} persistence diagrams of the *filtrations of the Vietoris-Rips complexes generated by X and Y* ,

$$d_B(D_X, D_Y) \leq d_{GH}(X, Y)$$

Bibliography

Some References:

♦ *Stability Results:*

- ❖ D. Cohen-Steiner, H. Edelsbrunner, J. Harer. ***Stability of persistence diagrams***. Discrete & Computational Geometry 37.1, pages 103-120, 2007.
- ❖ F. Chazal, D. Cohen-Steiner, M. Glisse, L. J. Guibas, S. Y. Oudot. ***Proximity of persistence modules and their diagrams***. Proc. of the 35 annual symposium on Computational Geometry, pages 237-246, 2009.
- ❖ F. Chazal, D. Cohen-Steiner, L. J. Guibas, F. Mémoi, S. Y. Oudot. ***Gromov-Hausdorff stable signatures for shapes using persistence***. Computer Graphics Forum 28.5, pages 1393-1403, 2009.